Amatya Sharma

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EDUCATION

University of Michigan, Ann Arbor Expected Apr 2027
Ph.D. Candidate, Computer Science and Engineering
GPA: 3.9/4 | Advisors: Nikhil Bansal and Euiwoong Lee

Indian Institute of Technology Kharagpur May 2022
Bachelor of Technology + Master of Technology, Computer Science
GPA: 9.69/10 | Department Rank 5 | University Rank 8

RESEARCH INTERESTS

Approximation, streaming, online, and parameterized algorithms; constraint satisfaction, clustering, and combinatorial optimization.

PUBLICATIONS AND MANUSCRIPTS

Authors are listed alphabetically by surname, following theory-CS convention.

Streaming Algorithms and Constraint Satisfaction

[1] Characterizing Turnstile Streaming Decidability of CSPs. Aditya Bandekar, Joshua Brakensiek, Aaron Putterman, Amatya Sharma, Santhoshini Velusamy. In submission to ACM Symposium on Theory of Computing (**STOC 2027**).

[2] Cores Characterize the One-Pass Streaming Complexity of CSPs. Joshua Brakensiek, Aaron Putterman, Amatya Sharma, Santhoshini Velusamy. Submitted to Innovations in Theoretical Computer Science (**ITCS 2027**).

[3] Non-Redundancy of Low-Arity Symmetric Boolean CSPs. Amatya Sharma, Santhoshini Velusamy. In submission to Symposium on Theoretical Aspects of Computer Science (**STACS 2027**).

[4] Characterizing Streaming Decidability of CSPs via Non-Redundancy. Amatya Sharma, Santhoshini Velusamy. European Symposium on Algorithms (**ESA 2026**). Invited to a special issue of **Theoretical Computer Science**.

Approximation Algorithms and Optimization

[5] Group-Based Rounding for FPT k-Median. Aditya Anand, Euiwoong Lee, Amatya Sharma. Submitted to Innovations in Theoretical Computer Science (**ITCS 2027**).

[6] Spectral Dual Fitting for k-Means. A. Anand, M. Charikar, V. Cohen-Addad, R. Gao, F. Grandoni, E. Lee, Amatya Sharma, E. van Wijland. ACM–SIAM Symposium on Discrete Algorithms (**SODA 2027**).

[7] Min-CSPs on Complete Instances II: Polylog Approximation for Min-NAE-3-SAT. Aditya Anand, Euiwoong Lee, Davide Mazzali, Amatya Sharma. Approximation Algorithms for Combinatorial Optimization Problems (**APPROX 2025**).

[8] Min-CSPs on Complete Instances. Aditya Anand, Euiwoong Lee, Amatya Sharma. ACM–SIAM Symposium on Discrete Algorithms (**SODA 2025**).

Online, Geometric, and Parameterized Algorithms

[9] Parameterized k-Core and Network Design Games. Amatya Sharma. In submission to Symposium on Theoretical Aspects of Computer Science (**STACS 2027**).

[10] Treewidth Parameterization of Network Design for Public Goods Games. Palash Dey, Amatya Sharma. Planned submission to International Joint Conference on Artificial Intelligence (**IJCAI 2027**).

[11] A Decomposition Approach to the Weighted k-Server Problem. Nikhil Ayyadevara, Ashish Chiplunkar, Amatya Sharma. Foundations of Software Technology and Theoretical Computer Science (**FSTTCS 2024**).

[12] Two-Dimensional Guillotine Strip Packing. Arindam Khan, Aditya Lonkar, Arnab Maiti, Amatya Sharma, Andreas Wiese. International Colloquium on Automata, Languages, and Programming (**ICALP 2022**).

[13] On Guillotine Separable Packings for the 2D Knapsack. Arindam Khan, Arnab Maiti, Amatya Sharma, Andreas Wiese. International Symposium on Computational Geometry (**SoCG 2021**). Contributed talk at Highlights of Algorithms (**HALG 2021**).

[14] On Parameterized Complexity of Liquid Democracy. Palash Dey, Arnab Maiti, Amatya Sharma. Algorithms and Discrete Applied Mathematics (**CALDAM 2021**).

RESEARCH AND PROFESSIONAL EXPERIENCE

Incoming Long-Term Visitor. Simons Institute for the Theory of Computing, UC Berkeley Jan–Apr 2027

- Will collaborate with leading researchers and pioneers in CSPs during *Symmetry in Efficient Computation with Local Constraints*.

Incoming Participant, Iliad Intensive. Iliad, Berkeley, CA Jan 4–29, 2027

- Will undertake intensive technical training in foundational AI alignment, including deep learning theory, agent foundations, and interpretability.

Research Visitor. TTIC; host: Santhoshini Velusamy Sep–Oct 2025

- Research focus: streaming complexity and constraint satisfaction problems.

Research Intern. IIT Delhi; host: Ashish Chiplunkar Apr–Oct 2021

- Conducted research on online algorithms.

Software Engineering Intern. Oracle R&D, ExaCC May–Jul 2021

- Implemented a Java library for secure HTTP authentication using the Java Cryptography Architecture.

Research Intern. IISc Bangalore; host: Arindam Khan May–Jul 2019; Dec 2019

- Conducted research in approximation algorithms and geometric packing.

TEACHING EXPERIENCE

Graduate Student Instructor. EECS 281, University of Michigan Jan–Apr 2026

Graduate Student Instructor. EECS 477, University of Michigan Aug–Dec 2024

Teaching Assistant. Algorithms 1, IIT Kharagpur Jan–Apr 2022

Teaching Assistant. Algorithmic Game Theory, IIT Kharagpur Aug–Dec 2021

ACADEMIC SERVICE AND OUTREACH

Reviewing. SODA 2027, SOSA 2027, Algorithmica 2026, RANDOM 2026, ICALP 2026, ITCS 2026, SODA 2026, APPROX 2024, ISAAC 2023, MFCS 2023.

Theory talks and lunches. Organized theory CS group talks and lunches, 2023–2024.

Annapurna. An initiative addressing hunger and food waste at IIT Kharagpur.

ACADEMIC ACHIEVEMENTS AND SCHOLARSHIPS

GATE Scholarship (2021). Teaching assistantship scholarship in CSE, IIT Kharagpur.

Department Change (2018). Transferred to CS after ranking among the top 10 of 1,300 first-year students at IIT Kharagpur.

JEE Advanced (2017). All India Rank 1,464 among approximately 170,000 students.

SJVN Merit Scholarship (2017). Awarded for performance in the CBSE Class 12 examination.

National Talent Search Examination Scholarship (2015). All India top 1,000 among approximately 600,000 students; State Rank 1, Himachal Pradesh.

GRANTS AND FUNDING

Rackham Conference Travel Funding. ESA 2026 ($1,150), APPROX 2025 ($900), FSTTCS 2024 ($1,400).

Additional conference and research travel support. APPROX 2025 conference grant ($200); NSF travel support for SODA 2025 ($700); DIMACS Fine-Grained Complexity Tutorial travel grant (2024).

SKILLS AND SELECTED COURSEWORK

Programming and tools. Python, C/C++, Java, Git, MySQL.

Algorithms coursework. Approximation and Online Algorithms, Parameterized Algorithms, Randomized Algorithms, Algorithmic Game Theory, Computational Complexity.

Machine learning coursework. Reinforcement Learning, Deep Learning, Advanced Machine Learning, Natural Language Processing, Statistical Foundations of AI and ML.